Too Much Information? Mandarin Classifiers from Humans to LLMs

Background. Surprisal theory assumes that processing difficulty tracks a word's information content, with higher surprisal implying greater processing difficulty (Eq. 1) [1]. Recent studies have shown that surprisal from **L**arge **L**anguage **M**odels (e.g., LLMs) often matches human reading times [2–3], raising the question of whether LLMs process language in ways that mirror human comprehension. Mandarin classifiers are a natural testbed for surprisal-based accounts and surprisal-distribution frameworks, such as Uniform Information Density (UID) [4], which predicts easier comprehension when surprisal is spread earlier rather than concentrated on a single word. Mandarin classifiers encode semantic and ontological features (e.g., animacy, shape) that constrain which nouns and adjectives can follow [5–6], where they **strongly constrain** nouns (e.g., **Grammatical**: yī tiáo$_{long}$ gǒu 'one CL$_{long}$ dog', **Ungrammatical**: *yī tiáo$_{long}$ shū 'one CL$_{long}$ book'), but the restriction between classifiers and adjectives is **more flexible** (e.g., yī tiáo$_{long}$ dà gǒu 'one CL$_{long}$ big dog', yī běn$_{flat}$ dà shū 'one CL$_{flat}$ big book'). Previous work shows that classifiers reduce processing difficulty at the following noun [7–9], but their effects on following adjectives remain unclear. We therefore test whether Mandarin classifiers support surprisal/UID predictions in both human processing and LLMs (e.g., Chinese BERT [10]).

Dataset and Design. For the human data, we use the Database of Eye-Movement Measures on Words in Chinese Reading [11], which provides word-level eye-tracking measures for 8k sentences. For LLM modeling, we use 19k sentences from the Chinese Classifier Dataset [12], and compute the word-level surprisal for both nouns and adjectives using Chinese BERT, constructing parallel **Numeral–(Classifier)–Adjective–De–Noun** sequences with and without classifiers via masked prediction (Fig. 1). For each target word, we compute a surprisal difference ($\Delta\text{surprisal}_{\text{target}}$, Eq. 2) by subtracting its surprisal in sentences ***with*** a classifier from its surprisal in sentences ***without*** a classifier, and link these values to human word-level eye-tracking data for the same nouns and adjectives, yielding 769 noun types and 678 adjective types matched across the two datasets.

Hypothesis. If surprisal follows UID, classifiers should reduce surprisal for upcoming nouns and adjectives in the LLM data (and plausibly in human processing). If LLM predictions mirror human comprehension, larger $\Delta\text{surprisal}_{\text{target}}$ should be associated with shorter reading times in naturalistic reading.

Results. Modeling results show that when a classifier is present, surprisal is reduced for both nouns and adjectives. **For nouns**, mean surprisal decreased from 9.46 without classifiers to 8.60 with classifiers (Δsurprisal = 0.86); **for adjectives**, surprisal dropped from 13.64 to 12.06 (Δsurprisal = 1.6), indicating stronger predictability benefits for adjectives than for nouns. Linking to human data, we correlated $\Delta\text{surprisal}_{\text{target}}$ with **log T**otal reading **T**ime (Mean_logTT: 2.58 **for nouns**, 2.59 **for adjectives**). The $\Delta\text{surprisal}_{\text{target}}$ was weakly negatively correlated with human log-TT for both adjectives and nouns, significant for adjectives but only marginal for nouns (**nouns**: $r(767) = -0.07$, $p = 0.06$; **adjectives**: $r(676) = -0.09$, $p < 0.05$) (Fig. 2).

Discussion. These results suggest that, consistent with UID principles, Mandarin classifiers act as general information-smoothing devices for LLMs, reliably reducing surprisal for both upcoming nouns and adjectives. In human processing, words that help the LLMs most are read slightly faster, but the correlations are weak. These findings show that any UID effect may be subtle, that LLM-based surprisal shows a weak alignment with human reading times, and that surprisal by itself **may not fully account for human sentence processing**.

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Eq. 1: Processing Difficulty ∝ surprisal(w) = −log P(w|c)

Eq. 2: Surprisal Reduction ∝ $\Delta\text{surprisal}_{target} = \text{surprisal}_{noClassifier}(w_{target}) - \text{surprisal}_{WithClassifier}(w_{target})$

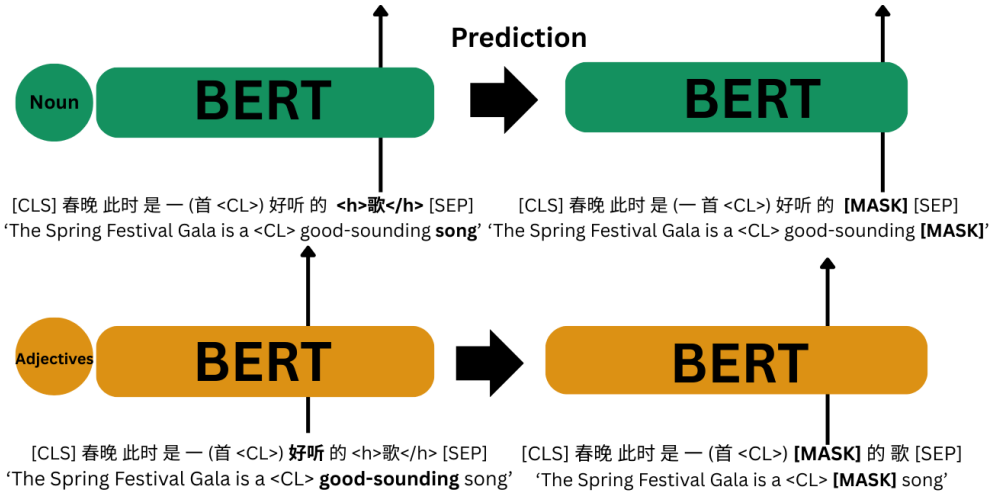

Fig 1: Sketch of our BERT-based noun and adjective selection models: predicting the noun by unmasking the [MASK] **(Top)**; predicting the adjective by unmasking the [MASK] **(Bottom)**

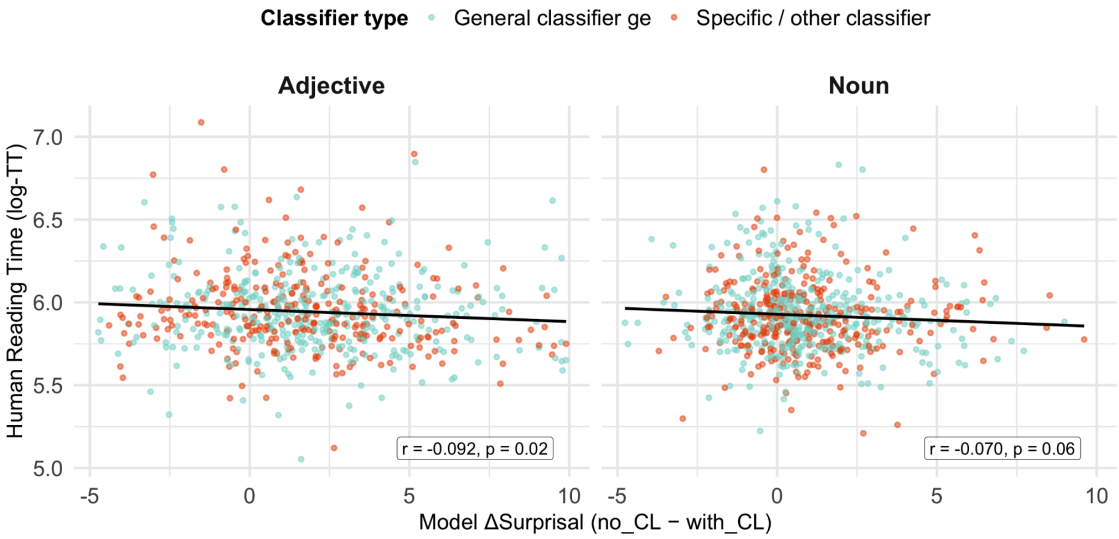

Fig 2: BERT Surprisal vs. Human Reading Times for Nouns and Adjectives (the general classifiers (e.g., ge), which can occur with all nouns, and specific classifiers, which only occur with specific noun classes (e.g., tiáo$_{long}$), were included in the analysis)

References: [1] Levy 2008 Cognition. [2] Merkx & Frank 2021 CMCL. [3] Wilcox et al. 2020 CSS. [4] Jaeger & Levy 2006 ANIPS. [5] Zhang 2013 Walter de Gruyter. [6] Chen et al. 2022 Cambridge handbook of Chinese linguistics. [7] Hao et al. 2025 JML. [8] Wu et al. 2009 WECOL. [9] Wu et al. 2011 Corpus Linguistics & Linguistic Theory. [10] Cui et al. 2021 IEEE/ACM. [11] Zhang et al. 2022 Scientific Data. [12] Peinelt et al. 2017 IJCNLP.

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